

Question		Answer	Mark	Mark scheme	Additional guidance
16	(a)	−6	B1	cao	For reference $f(x) = \frac{12}{x+1}$ and $g(x) = 5 - 3x$ Accept $\frac{4}{1}$ Could be shown in the form of a flowchart, which must show correct inverse operations. Allow g^{-1} and g to be in terms of y ie accept $\frac{5-y}{3}$ and $5 - 3y = 4$
	(b)	4	M1	for $g(1) = 5 - 3 \times 1 (= 2)$ and a clear intention to find $f(“2”)$ or for $\frac{12}{5-3 \times 1+1}$ or for stating fg, eg $\frac{12}{5-3x+1}$ oe	
	(c)	$\frac{1}{3}$	A1	cao	
			M1	for g^{-1} as $\frac{5-x}{3}$ oe or for $5 - 3x = 4$ or $g\left(\frac{1}{3}\right) = 5 - 1 = 4$	
			A1	for $\frac{1}{3}$ oe eg 0.33(3...)	